

14	(a)	reduction in power	(allow intensity/amplitude)	B1	[1]
	(b)	(i)	attenuation = 2.4×30 = 72 dB	A1	[1]
		(ii)	gain/attenuation/dB = $10 \lg(P_2/P_1)$	C1	
			$72 = 10 \lg(P_{\text{IN}}/P_{\text{OUT}})$ or $-72 = 10 \lg(P_{\text{OUT}}/P_{\text{IN}})$	C1	
			ratio = 1.6×10^7	A1	[3]
	(c)	e.g. enables smaller/more manageable numbers to be used e.g. gains in dB for series amplifiers are added, not multiplied		B1	[1]